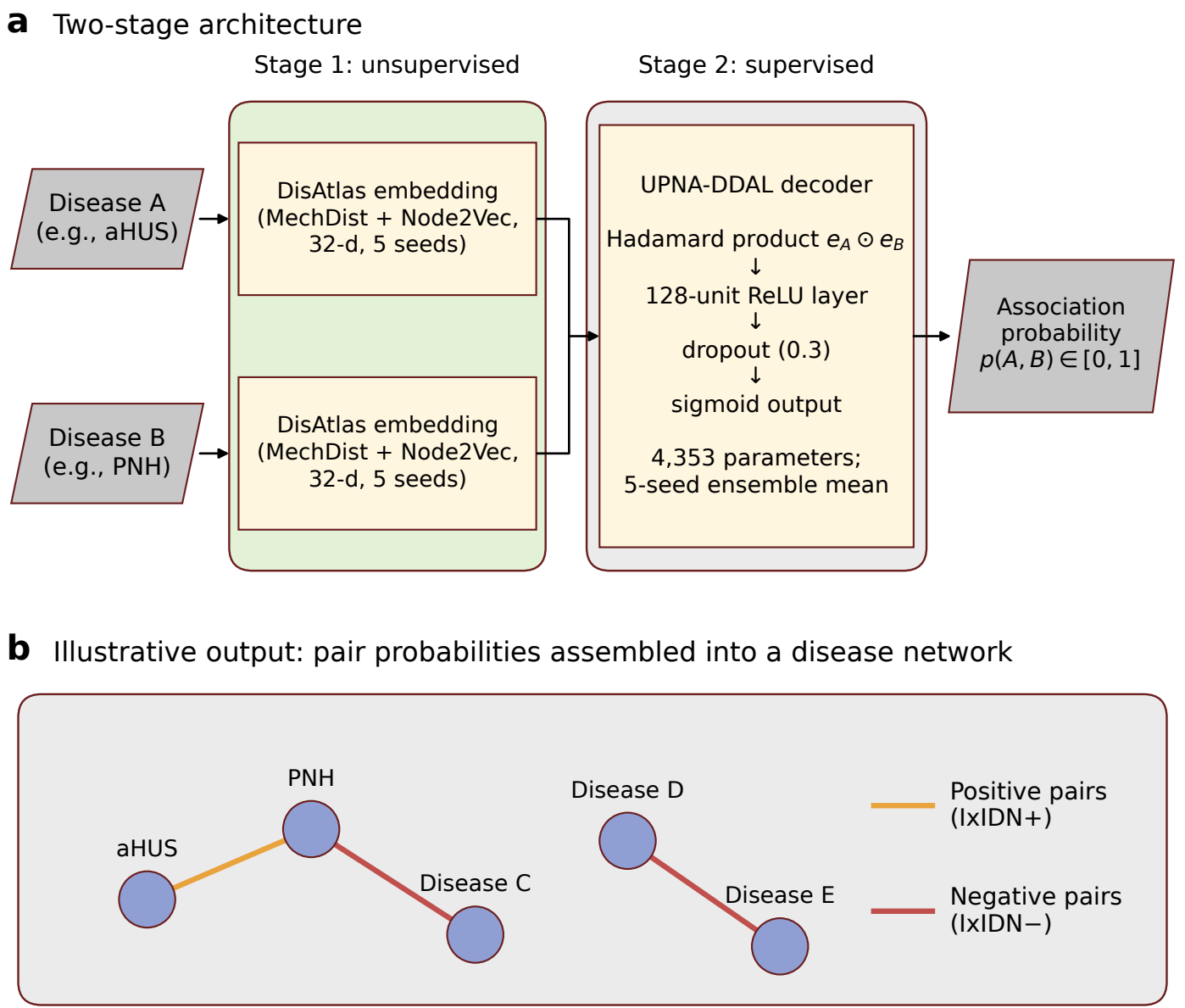
a Two-stage architecture
Stage 1: unsupervised
Stage 2: supervised
Disease A (e.g., aHUS)
Disease B (e.g., PNH)
DisAtlas embedding (MechDist + Node2Vec, 32-d, 5 seeds)
DisAtlas embedding (MechDist + Node2Vec, 32-d, 5 seeds)
UPNA-DDAL decoder
Hadamard product $e_A \odot e_B$
↓
128-unit ReLU layer
↓
dropout (0.3)
↓
sigmoid output
4,353 parameters;
5-seed ensemble mean
Association probability $p(A, B) \in [0, 1]$
b Illustrative output: pair probabilities assembled into a disease network
aHUS
PNH
Disease C
Disease D
Disease E
Positive pairs (IxIDN+)
Negative pairs (IxIDN−)